Research Assignment 2 27 April 2026

SQL Aggregate Functions & Operators

1. Select distinct department
from students;
Output

Departments
IT
HR
Finance

2. Select department, AVG(age) AS avg_age
from Students
Group by department;
Output

Department	Avg_age
IT	20,5
HR	22
Finance	23

3. Select department, Count(*) AS student_count
From Students
Group by department
Having count(*) > 1;
Output

Department	Student_count
IT	2
HR	2

4. ~~Stude~~

Select Student_id, name, age, department
From Students
Wher age between 21 and 23;

Output

student id	Name	Age	Department
2	Bob	22	HR
3	Charlie	21	IT
4	Diana	23	Finance
5	Eve	22	HR

5. Select student_id, name, age, department
From students
Where department IN ('IT', 'HR')
AND age > 21;

Output

Student_id	Name	Age	Department
2	Bob	22	HR
5	Eve	22	HR

6. Select department, Sum(credits) as total_credits
From courses
Group by departments
Having Sum(credits) > 5;

Output

Department	Total_credits
IT	11

7. Select course_id, course_name, department, credits
From courses
where credits <> 4;

Output

Course_id	Course_name	Department	credits
101	SQL Basics	IT	3
104	Excel	Finance	2
105	Statistics	HR	3

8. Select course_id, course_name, credits
From courses
Order by credits descending
Limit 3;

Output

Course_id	Course_name	credits
102	Python	4
103	Data Science	4
101	SQL Basics	5

9. Select
Max(grade) As max_grade,
Min(grade) As min_grade,
AVG(grade) As avg_grade,
From enrollments;

Output

Max_grade	Min_grade	Avg_grade
90	78	84,6

10. Select course_id,
Count(*) as enrollment_count
From enrollments
Group by course_id;

<u>Output</u>

Course_id	Enrollment_count
101	1
102	1
103	1
104	1
105	1

11. Select department,
Sum(salary) as total_salary,
Sum(bonus) as total_bonus
From salaries
Group by department;

<u>Output</u>

Department	Total_Salary	Total_bonus
IT	122 000	10 500
HR	109 000	7500
Finance	70 000	6000

12. Select department,
Avg(salary) as avg_salary
From salaries
Group by department
Having AVG(salary) > 55 000;

Output

Department	Avg_Salary
IT	61 000
Finance	70 000

13. Select employee_id,
name,
Salary,
bonus,
(salary + bonus) as total_compensation
From Salaries
Where (salary + bonus) > 60 000;

Output

Employee_id	Name	Salary	Bonus
1	Tom	60 000	5000
3	Spike	70 000	6 000
4	Tyke	62 000	5 500

14. Select department,
Sum(budget) as total_budget,
Avg(budget as avg_budget.
From projects
Group by department
Having Avg(budget) > 70 000;

Output

Department	Total_budget	Avg_budget
IT	270 000	135 000
Finance	80 000	80 000

15. Select project_id,
Project_name
Department,
Budget
From projects
where budget between 50000 and 120 000
And department <> 'Marketing';

Output

Project_id	Project_name	Department	Budget
1	AI App	IT	120 000
2	Payroll System	Finance	80 000
5	HR Portal	HR	50 000